

SIMULINK REPORT

This report presents the simulations of feedback control methods for a motor speed system. The figures of Chapter 1 and 2 are recreated and simulated using MATLAB/Simulink. The system behaviours obtained are explained and commented on.

Simulink Model Figure 1.5:

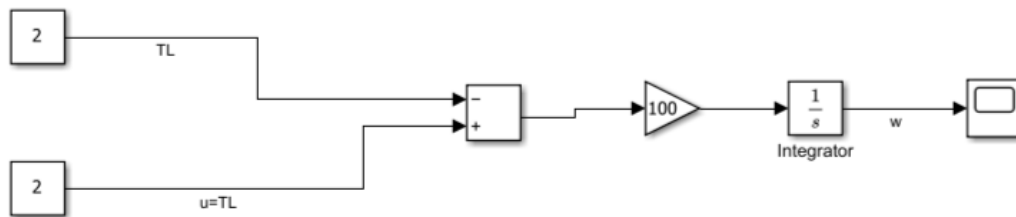


Figure 1.5.1: Simulation diagram of the open loop feedforward control for $T_L = 2 \text{ Nm}$ and $J = 0.01 \text{ kg m}^2$.

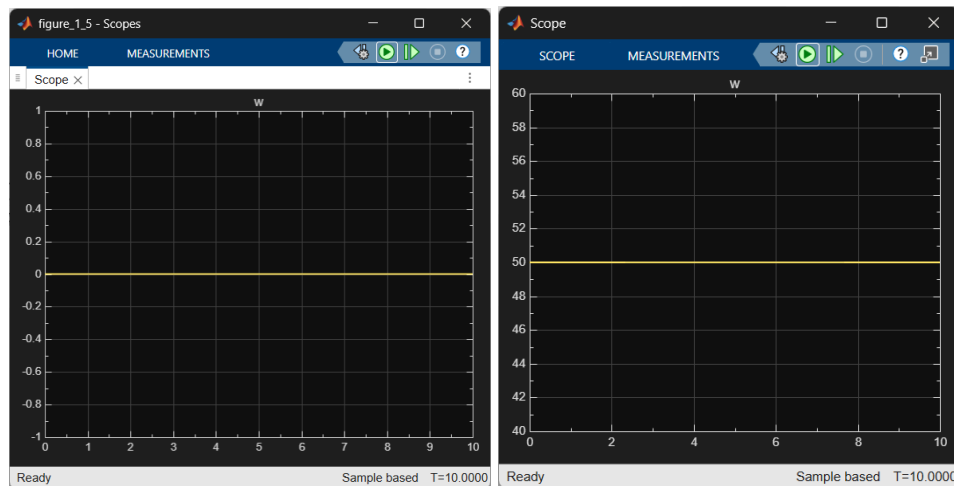


Figure 1.5.2: Speed response of the feedforward control for $\omega(0) = 0$ and $\omega(0) = 50$, respectively

In this open-loop scheme, the controller only applies a constant torque based on the estimated load ($T_L = 2$). The system cannot correct any external errors or initial condition mismatches. If the motor's initial speed is exactly 50 rad/s, it stays at 50. However, if the initial speed is different (for example, $\omega(0) = 0$), the speed just stays at 0. It never reaches the 50 rad/s reference because there is no feedback to tell the controller that the speed is wrong.

In conclusion, this simulation illustrates that the open-loop controller only provides the exact torque required to overcome the load, under the assumption that the motor is already operating at the desired speed.

Simulink Model Figure 1.6:

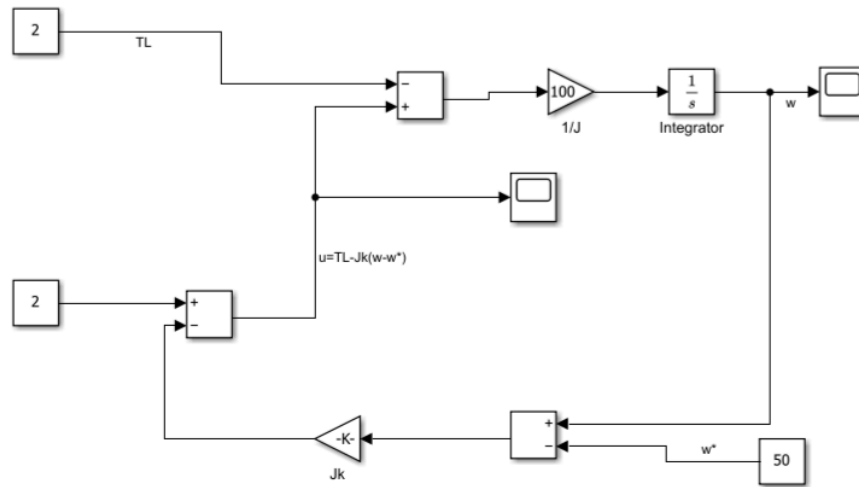


Figure 1.6.1: Simulation diagram of the closed-loop proportional control for $T_L = 2 \text{ Nm}$ and $J = 0.01 \text{ kg m}^2$ and $k = 5$

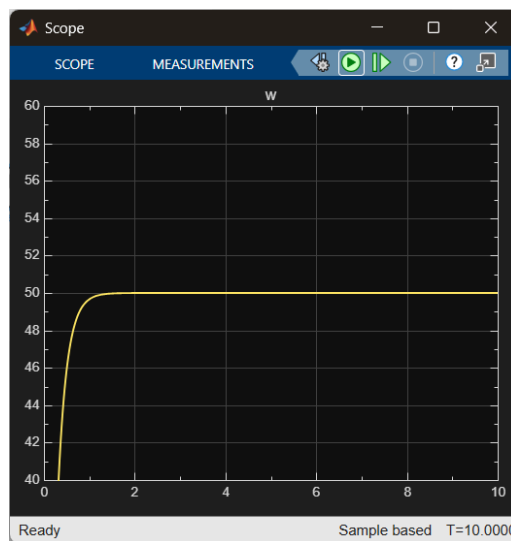


Figure 1.6.2: The speed response of the proportional control for the initial condition of $\omega(0) = 0$

A proportional feedback control is introduced to correct the limitations of the only feedforward control. This is obtained in the model by calculating the tracking error by subtracting the reference constant ($\omega^* = 50$) from the actual output speed (ω). A proportional gain block ($Jk = 0.05$) is then used to scale this error. Then, this term is subtracted from the nominal load torque ($T_L = 2$), creating the closed-loop control: $u = T_L - Jk(\omega - \omega^*)$.

The success of this method is due to the stability of the error dynamics. As seen in the result for the initial condition $\omega(0) = 0$, the system begins with a maximum tracking error

($\epsilon(t) = \omega^* - \omega(t) = 50 - 0 = 50$). The closed loop senses this error and produces a corrective torque proportional to the error. As the motor speeds up, the error becomes smaller. As a result, the motor speed exponentially converges to the reference value of 50 rad/s.

This simulation demonstrates how adding the proportional feedback term solves the problem of the initial condition of the system, which is the limitation of the open loop control.

Simulink Model Figure 1.7:

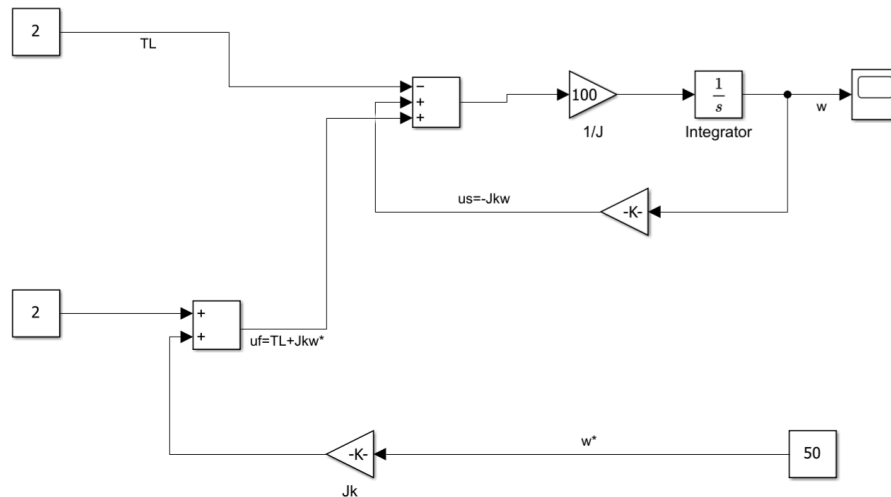


Figure 1.7.1: Simulation diagram of the virtual friction perspective of proportional control for $T_L = 2 \text{ Nm}$ and $J = 0.01 \text{ kg m}^2$ and $k = 5$

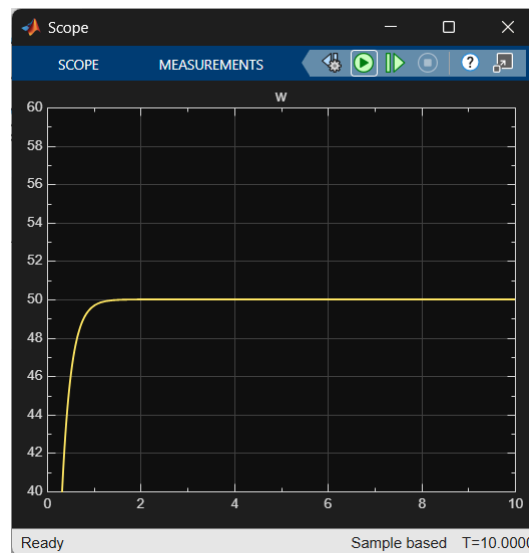


Figure 1.7.2: Speed response for the initial condition $\omega(0) = 0$

Figure 1.7 illustrates a representation of the proportional controller in Figure 1.6 with a new physical insight into the identical mathematical idea. The block responsible for computing the error (by subtracting the actual speed from the desired one) is eliminated. The proportional

control is decomposed into two independent terms, which are the feedforward term ($u_f = T_L + Jk\omega$) and the state feedback stabilizing term ($u_s = -Jk\omega$).

The feedback term ($u_s = -Jk\omega$) acts as a virtual friction that is inserted directly into the motor dynamics. This virtual friction makes changes to the physical properties of the motor in order to maintain the asymptotic stability of the error dynamics. Meanwhile, the u_f term acts as a constant driving force. Since the u_s acts like friction and keeps the motor stable, u_f provides the exact power needed to drive the motor to the desired speed of 50 rad/s.

As seen in the scope, the speed response for $\omega(0) = 0$ is the same as the graph of Figure 1.6. This shows that the general control law is unchanged ($u = u_f + u_s$). Therefore, it can be said that mathematically rearranging the controller provides a viewpoint on how feedback significantly changes a dynamic system without changing the basic closed-loop behavior.

Simulink Model Figure 1.8:

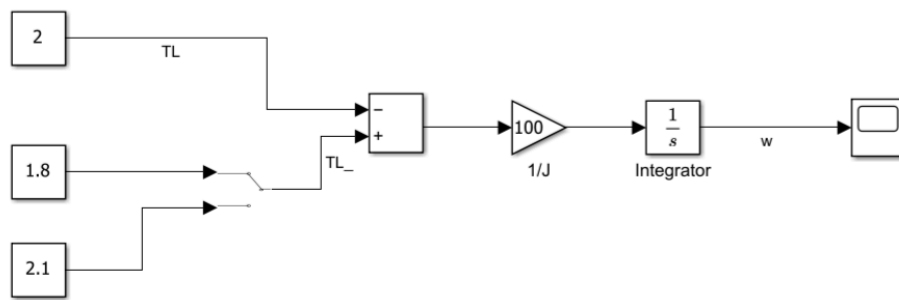


Figure 1.8.1: Simulation diagram of the feedforward control with incorrect load torque compensation (estimated load $\overline{T}_L = 1.8$ and $\overline{T}_L = 2.1$ Nm)

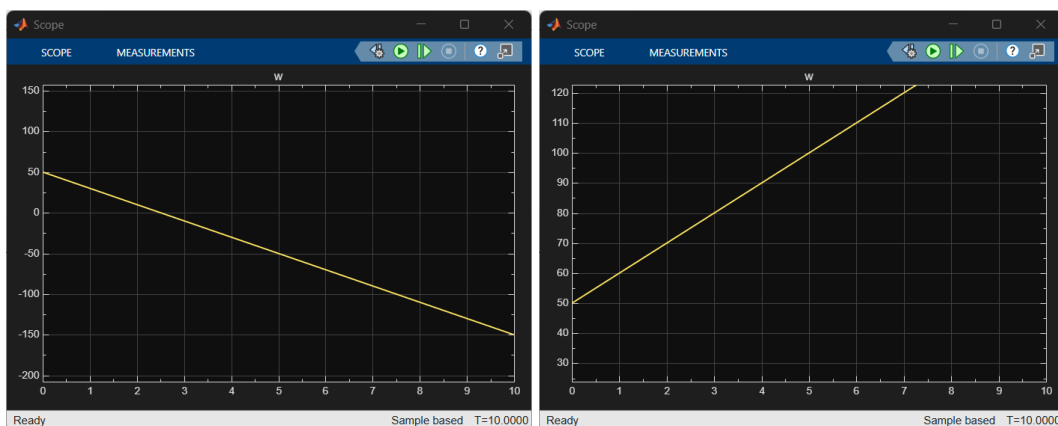


Figure 1.8.2: Speed responses (for $\overline{T}_L = 1.8$ and $\overline{T}_L = 2.1$ respectively)

In this model, the behavior of feedforward control is observed when the load torque is estimated incorrectly. Since the exact load is not always known in practical applications, the

actual motor load is kept constant at $T_L = 2$ Nm, but incorrect load estimates ($\overline{T}_L = 1.8$ Nm and $\overline{T}_L = 2.1$ Nm) are given to the controller using a switch.

The scope shows that even if the motor is started perfectly at the target speed ($\omega(0) = 50$), this speed cannot be maintained. When the load is underestimated (1.8 Nm), enough torque is not generated, causing the motor to slowly stop (decelerates). On the other hand, when the load is overestimated (2.1 Nm), too much power is applied and the motor speeds up continuously (accelerates).

Since there is no feedback loop to check the actual speed and fix the error, even a small mistake creates a constant extra torque on the motor. Since the system is not instructed to correct itself, this extra torque pushes the motor speed away from the 50 rad/s target.

Simulink Model Figure 1.9:

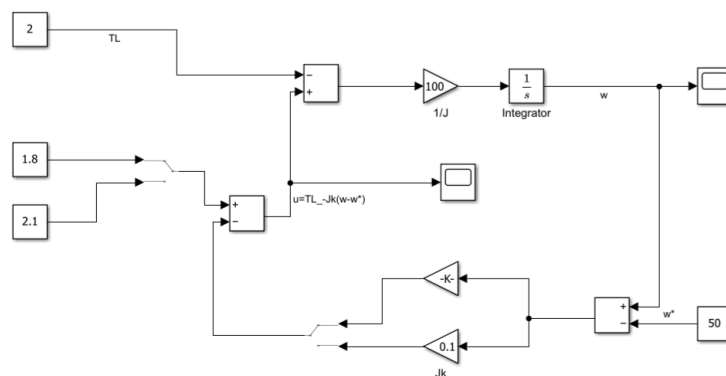


Figure 1.9.1: Simulation diagram of the proportional control under incorrect load torque compensation ($\overline{T}_L = 1.8$ and $\overline{T}_L = 2.1$) and different proportional gain ($k = 5$ and $k = 10$)

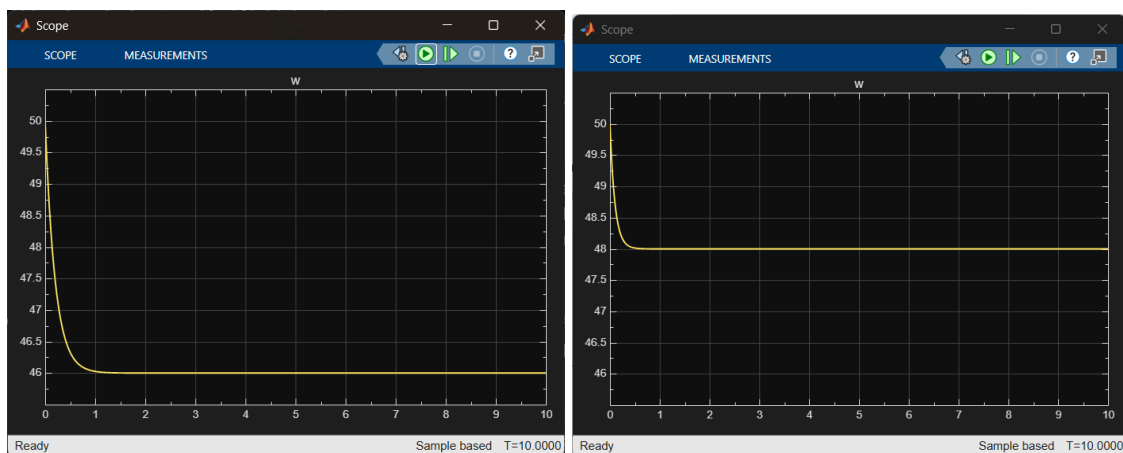


Figure 1.9.2: Speed responses showing the steady-state error and the effect of increasing the proportional gain ($k = 5$ and $k = 10$)

In this model, the closed-loop proportional control is tested under the same incorrect load estimation problem seen in Figure 1.8. The actual load is $T_L = 2$ Nm, but the controller is fed

an underestimated value of $\overline{T}_L = 1.8 \text{ Nm}$. As observed in the scopes, the motor speed initially drops from the 50 rad/s, but it does not drift away forever like it did in the open-loop system. Instead, the speed is stabilized and settles at a constant value slightly below the reference. This stabilization shows the advantage of the feedback loop. When the speed drops, the proportional controller detects the error and generates extra torque to compensate for the missing 0.2 Nm. However, to produce this continuous corrective torque, the proportional controller requires a continuous error to exist. Therefore, the motor speed can never perfectly reach the 50 rad/s. This remaining gap is known as the steady-state error. By comparing the two scope results, it is clearly seen that when the proportional gain (k) is increased from 5 to 10, the steady-state error becomes smaller and the speed gets closer to the reference. However, it is never completely eliminated using only proportional control.

Simulink Model Figure 1.11:

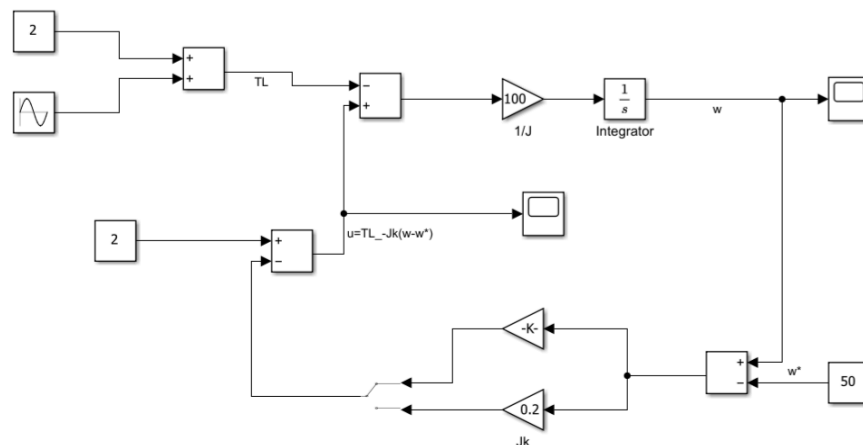


Figure 1.11.1: Simulation diagram of the proportional control under a time-varying load torque compensation ($T_L = 2 + 0.2\sin(12t)$)



Figure 1.11.2: Speed responses showing the effect of different proportional gains ($k = 5$ vs $k = 20$) on load fluctuations

In this simulation, the closed-loop system is subjected to a time-varying, sinusoidal load torque instead of a constant error. This scenario represents real-world conditions where the load fluctuates dynamically (such as varying friction or wind loads). The actual load is modeled using a Sine Wave block as $T_L = 2 + 0.2\sin(12t)$, while the controller is still only designed to compensate for the actual $T_L = 2$ Nm.

As seen in the scope results, the motor speed no longer settles at a perfectly flat line. Instead, the speed output continuously oscillates at the same frequency as the load disturbance (12 rad/s).

However, when the proportional gain is increased from 5 to 20, the amplitude of the unwanted speed oscillations is reduced. This clearly demonstrates another advantage of the feedback loop which is disturbance attenuation. A higher proportional gain forces the controller to react more aggressively to the load fluctuations. Thus, the disturbance is suppressed and the motor speed is kept much closer to the desired stable state.

Simulink Model Figure 1.12:

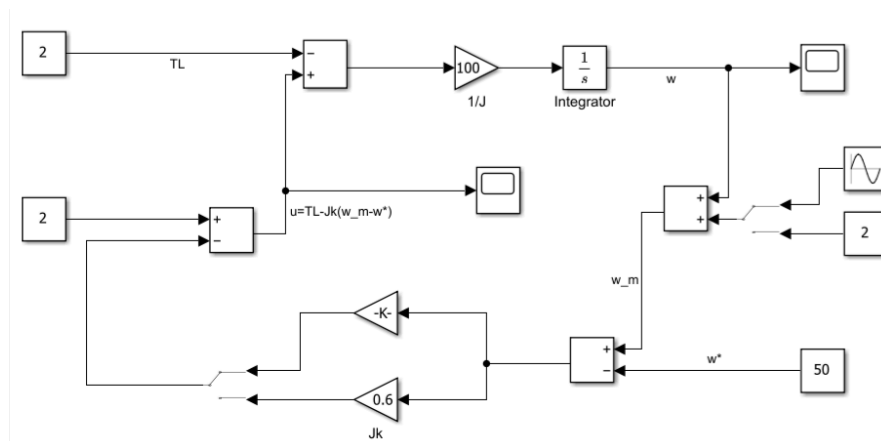


Figure 1.12.1: Simulation diagram of the proportional control with measurement noise ($d_m = 0.3\sin(24t)$)

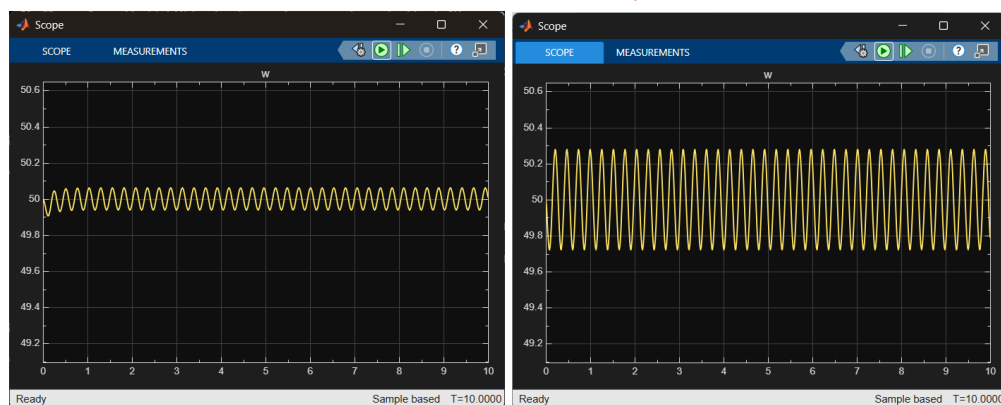


Figure 1.12.2: Speed responses showing the amplification of high frequency sensor noise for low ($k=5$) and high ($k=60$) proportional gains

In this model, measurement inaccuracies are added to the feedback loop. It includes a manual switch to test two different sensor error scenarios: a calibration error (Constant=2) and a high frequency measurement noise ($d_m = 0.3\sin(24t)$).

If the constant error scenario (constant = 2) is tested, it simulates a sensor that always measures the speed 2 rad/s higher than the real value. As a result, the motor's actual speed is automatically reduced by the controller to fix this fake error, leaving the final speed permanently below the 50 rad/s target.

The scenario presented in the scope results is the measurement noise. As seen in the model, this sinusoidal noise directly enters the closed loop system and is multiplied by the proportional gain block. In Figure 1.11, it was determined that increasing the gain helps to reduce load fluctuations. But this simulation highlights the disadvantages of having a large gain.

When the gain is kept low at $k=5$, the motor speed shows small, tolerable vibrations. But when the gain is aggressively increased to $k=60$, the sensor noise is increased by the controller. As a result, the motor is forced to respond to fake, quickly changing speed mistakes, which results in rapid oscillations in the motor's real speed.

Although a large proportional gain makes the system very sensitive to sensor noise, it is good for avoiding external load disturbances. As a result, the gain needs to be properly adjusted.

Simulink Model Figure 2.5:

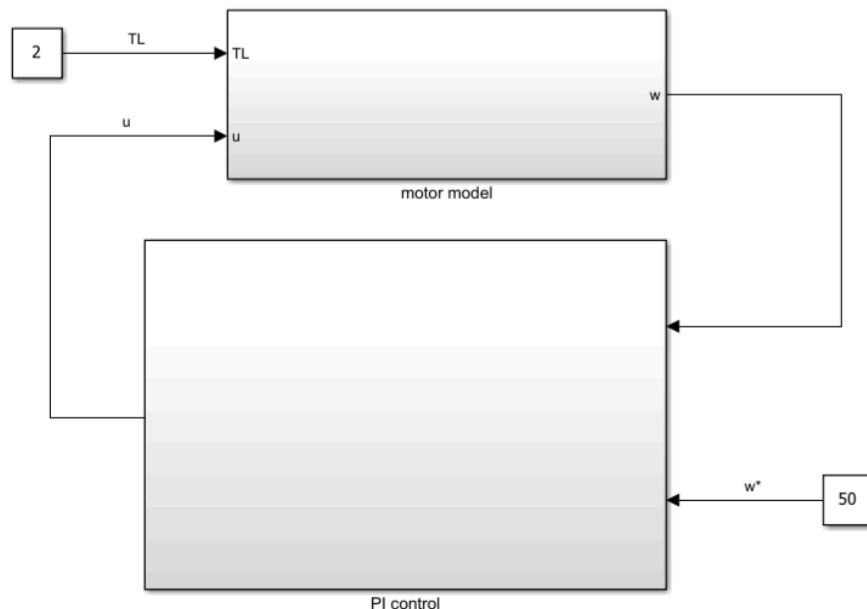


Figure 2.5.1: Dynamic system (PI control) controlling a dynamic system (electrical motor)

Two subsystems are used to represent the system in Simulink Model Figure 2.6: the motor model and the proportional integral controller. The illustration of the structure is important since it shows that the controller is currently a dynamic system rather than just a gain.

Simulink Model Figure 2.6:

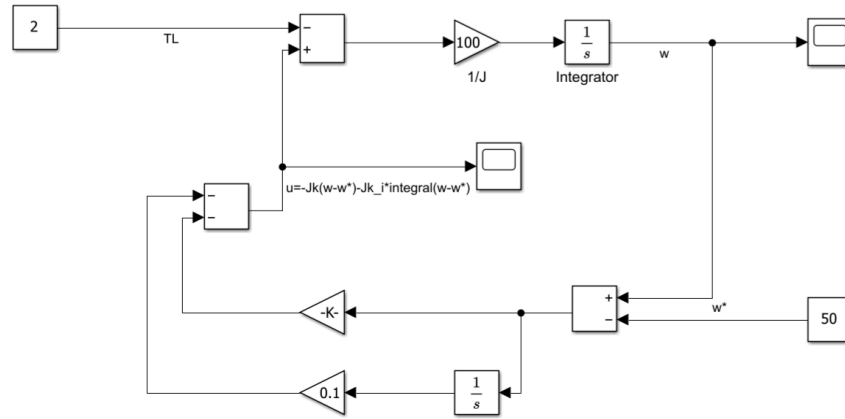


Figure 2.6.1: Simulation diagram of the proportional integral (PI) control system

The controller has been transformed to a PI controller in this case. Unlike the approach in Chapter 1, the controller design does not include any feedforward term ($\overline{T_L}$). As a result, there is no information about the load's actual value ($T_L = 2 \text{ Nm}$) and the controller relies entirely on the feedback error ($\omega - \omega^*$) through the proportional and integral paths.

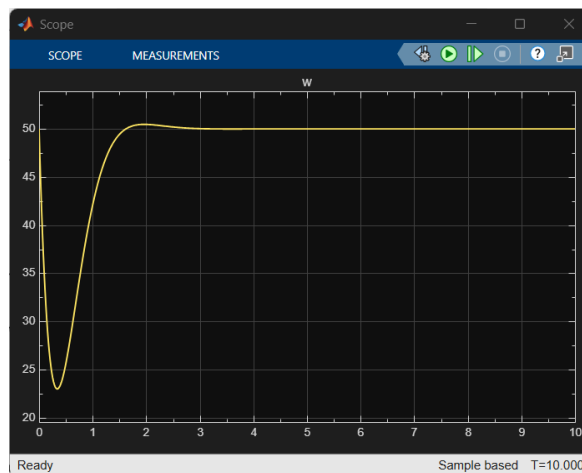


Figure 2.6.2: Speed response illustrating the elimination of steady-state error using PI gains $k = 5$ and $k_I = 10$

For the nominal case, the gains are set to $k = 5$ and $k_I = 10$. The addition of the integral term changes the behavior of the system, as seen in Figure 2.6.2. The motor speed maintains the 50 rad/s reference, and the steady-state error is completely gone. This is because the integral block integrates the speed error continuously and provides the precise control action needed to cancel out the disturbance of 2 Nm.

The system is then analyzed under three distinct gain sets in order to fully understand the influence of the proportional and integral actions. These combinations directly relate to the explicit mathematical solutions of the closed-loop system's characteristic equation:

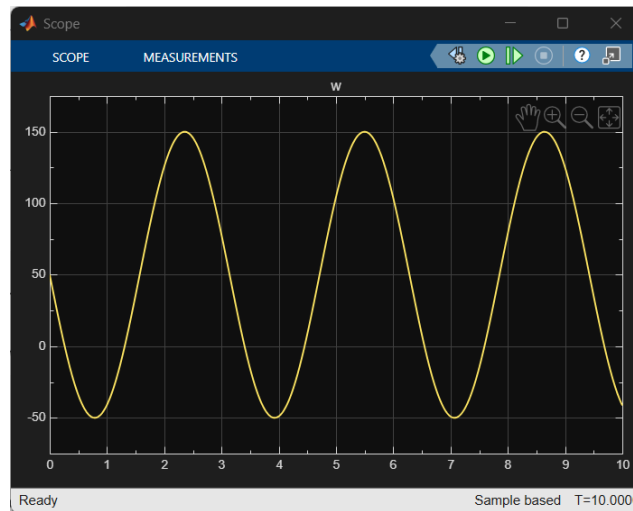
$$s^2 + ks + k_I = 0.$$


Figure 2.6.3: Speed response with the PI gains $k = 0$ and $k_I = 4$ (only integral control)

The characteristic equation becomes $s^2 + 4 = 0$ when the proportional gain is zero, producing only imaginary roots ($s = \pm 2j$). This solution represents an undamped oscillator. The scope shows that the motor speed never gets fixed and continuously oscillates as a sine wave around the reference. All necessary damping is gone from the system when the proportional term is absent.

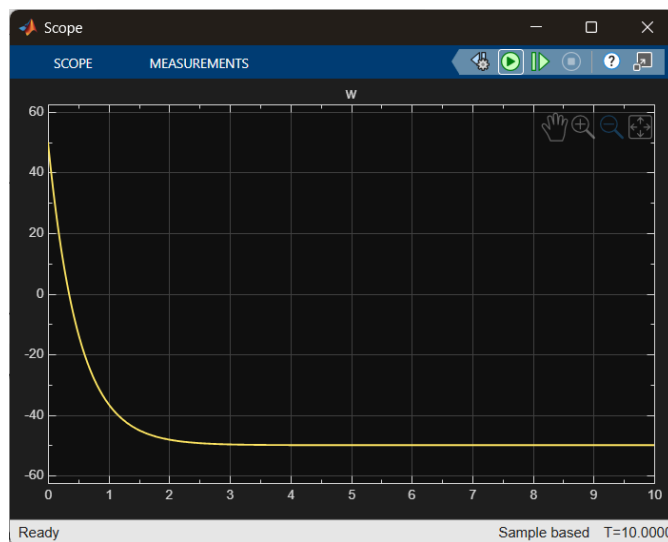


Figure 2.6.4: Speed response with the PI gains $k = 2$ and $k_I = 0$ (only proportional control)

When the integral gain is set to zero, the system becomes the same as the proportional control which excludes the integral control's effect and thus, the speed drops to a stable

value of less than 50 rad/s. The lack of an integral effect prevents the system from learning the unknown load, which leads to a steady state error.

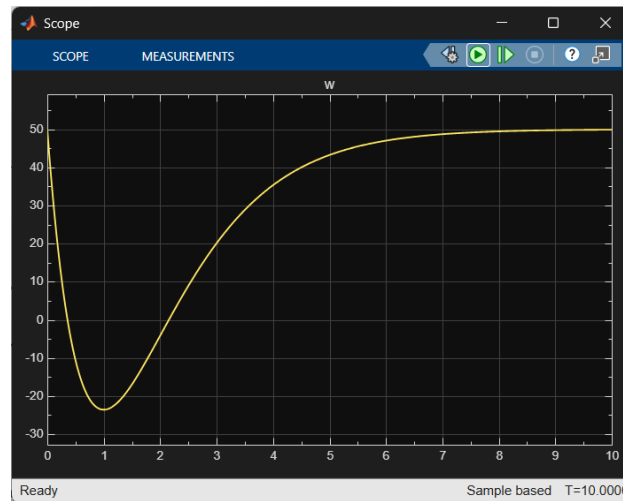


Figure 2.6.4: Speed response with the PI gains $k = 2$ and $k_i = 1$ (critically damped PI control)

In this case, the characteristic equation becomes $s^2 + 2s + 1 = 0$, which has real and repeated roots ($s = -1, -1$). This represents a critically damped system. The motor speed smoothly recovers from the load disturbance and converges to the 50 rad/s reference perfectly.

In conclusion, this model illustrates the roles of the controller parameters. While the integral gain (k_i) provides the memory needed to remove the steady-state error, the proportional gain (k) creates damping and temporary stability. A smooth and accurate reaction requires a carefully balanced combination, as seen in Case 3.

Simulink Model Figure 2.7:

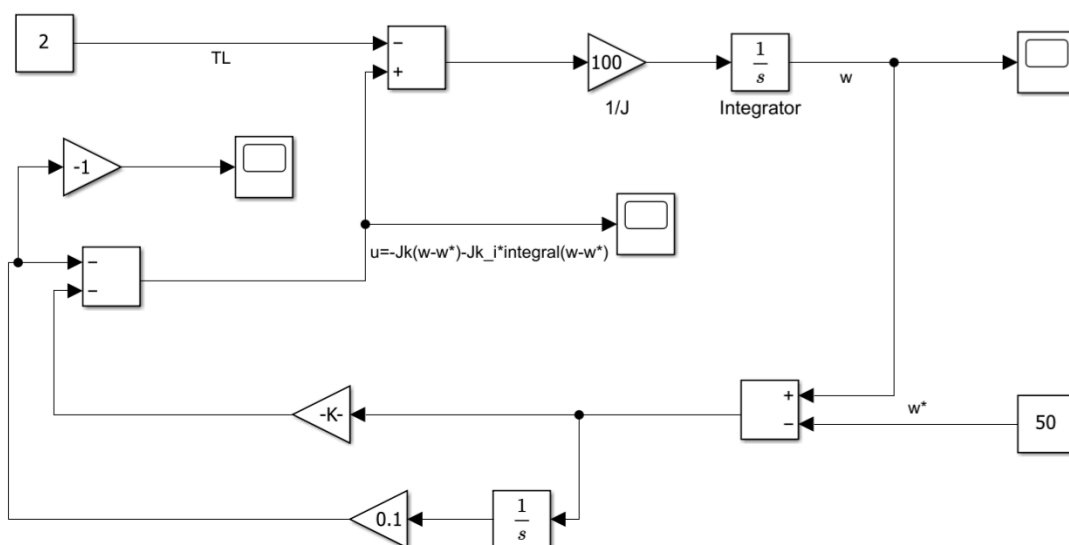


Figure 2.7.1: Simulation diagram of proportional integral control (estimation properties)

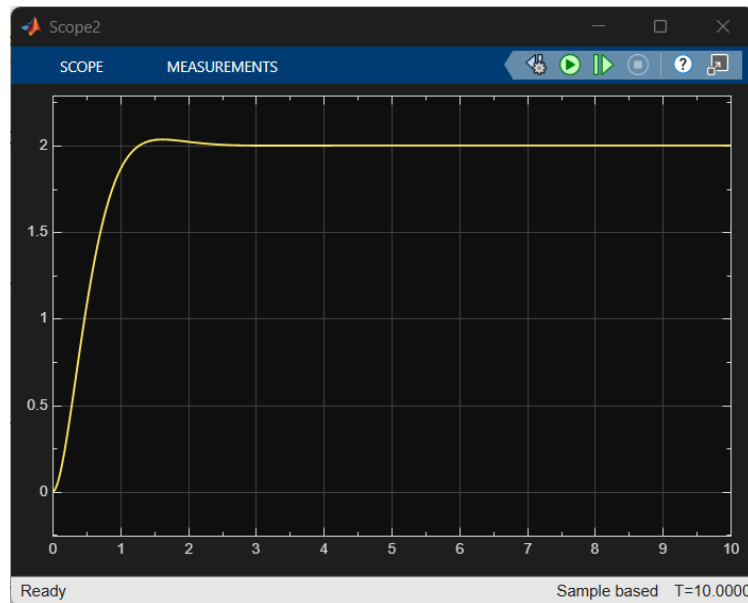


Figure 2.7.1: Time profile of the integral term $v(t)$ demonstrating the autonomous estimation of the load torque

The time profile of the integral term, $v(t)$, is analyzed to confirm how the PI controller dynamically estimates an unknown constant disturbance. In control theory, system errors are perfectly canceled by the integral action, which autonomously reaches a steady-state value. The motor undergoes a negative disturbance ($-T_L$) due to the physical load. The integral term must produce an equal and opposite control action in order to achieve zero steady-state error without any prior feedforward load estimation ($\overline{T_L} = 0$). To fully negate the $-T_L$ disturbance, the integral term must converge to $+T_L$. The integral signal $v(t)$ begins at zero and converges to $+2$ Nm, matching the load, as shown by the scope. By automatically correcting for the disturbance and enabling the motor speed to lock onto the reference without any steady state error, this demonstrates that the integral action operates as a perfect dynamic estimator.

To conclude, the simulations highlight the difference between integral and proportional controls. Although the Proportional (P) controller offers system stability, when an external load disturbance occurs, it leaves a steady state error. Meanwhile, the Proportional Integral (PI) controller overcomes this restriction by adding the integral term. By eliminating the unknown load torque, the integral action functions as an independent dynamic estimator, bringing the steady state error to zero and guaranteeing accurate speed tracking.